Remote Sensing for Forest Health and Wildfire Mitigation

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Introduction

Forest ecosystems play a vital role in biodiversity, soil stability, and water regulation but face increasing threats from wildfires and drought. Having personally experienced wildfires in the Los Angeles National Forest, there is an urgent need for predictive tools to mitigate their impact. This research utilizes remote sensing to assess forest health and wildfire risk by analyzing drought severity and vegetation indices in Humboldt County. The goal is to develop a geospatial framework for identifying high-risk areas and informing sustainable forest management strategies.

Humboldt County (Figure 1) is located in the northwestern region of California, characterized by its rugged coastline, extensive redwood forests, and temperate maritime climate. The county experiences cool, wet winters and dry summers, with significant precipitation contributing to high forest biomass. The dominant vegetation includes coast redwoods (*Sequoia sempervirens*), Douglas fir (*Pseudotsuga menziesii*), and tanoak (*Notholithocarpus densiflorus*), forming dense, old-growth forests.



Figure 1: Map of the State of California highlighting the location of Humboldt County.

Objective

The primary objective of this research is to develop a framework for using geospatial data to proactively manage wildfire risks and improve forest health. This study explores how remote sensing can be used to:

- Quantify forest health
- Assess spatial and temporal patterns of drought and vegetation indices
- Identify areas at high risk for wildfires

Methods

Moderate Resolution Imaging Spectroradiometer (MODIS)

Provides vegetation health metrics such as the Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI.

Spatial resolutions range from 250 m ters to 1 km

Landsat 8 8-9 Operational Land Imager (OLI)

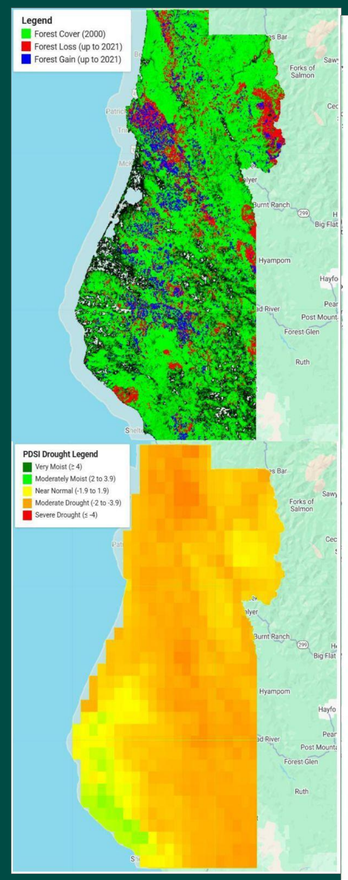
Captures multispectral imagery at 30-meter resolution

Enables detailed analysis of forest cover changes, drought stress, and post-fire recovery

Gridded Meteorological Data (GRIDMET)

Supplies high-resolution climate variables, including precipitation, temperature, andthe Palmer Drought Severity Index (PDSI)

Contextualizes vegetation trends and assess drought impacts on forest health



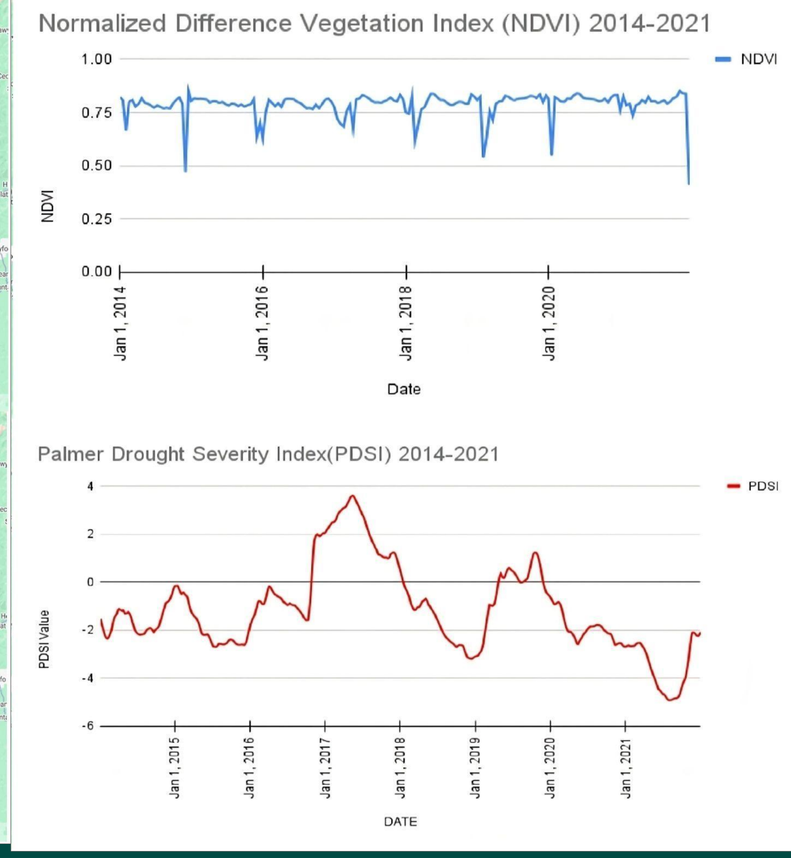


Figure 2: Vegetation Health: MODIS NDVI values (0.6–0.8) suggest overall forest vitality in Humboldt County.
Climatic Trends: GRIDMET data, particularly PDSI, indicates prolonged drought severity, which may contribute to forest stress despite positive NDVI values.

Results

- Forest Loss vs. Growth: Landsat imagery confirms forest loss is outpacing regeneration, likely due to drought stress, wildfires, or both (Figure 2).
- Environmental Pressures: While MODIS provides a vegetation snapshot, integrating GRIDMET and Landsat reveals underlying factors weakening forest resilience (Figure 2).
- Remote Sensing Approach: Findings emphasize the necessity of a multi-scale analysis to monitor vegetation health, assess climate-related stressors, and improve wildfire risk assessment.

Next Steps

To further develop this research and achieve the objective of identifying high-risk wildfire areas, the following next steps will be undertaken:

- Incorporate Additional Datasets: Integrate fire history data from sources like the Monitoring Trends in Burn Severity (MTBS) dataset and Cal Fire incident reports to refine risk assessment models.
- Machine Learning Applications: Explore predictive modeling approaches using machine learning algorithms to classify fire-prone zones based on vegetation health and climate trends.

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We would like to thank McNair Scholars Program, COMPASS, and the Alfred P. Sloan Foundation for supporting this work.